F FWallSim · Firewall Course

CLASS 5

Publish a web server with a Virtual IP

Destination NAT (VIP), inbound policy

Beginner | ~20 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

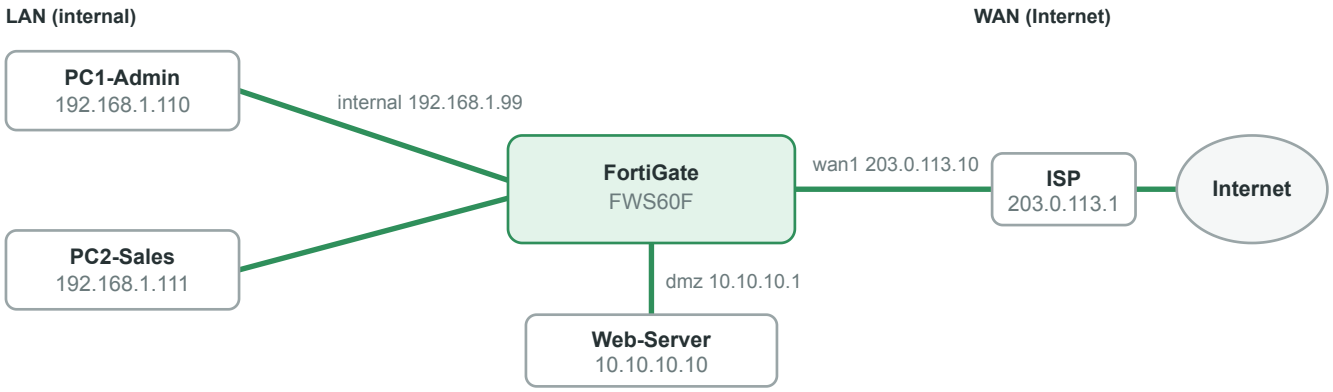
YOU WILL LEARN

- Destination NAT (VIP)
- inbound policy

BEFORE YOU START

Finish Class 4 first.

The lab network



The goal

THE IDEA

The company website runs on Web-Server (10.10.10.10, DMZ). Make it reachable from the internet on the wan1 address, for HTTP only.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **Publish a web server with a Virtual IP** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **Create a VIP: wan1 203.0.113.10 TCP/80 → 10.10.10.10:80**

   Policy & Objects > Virtual IPs > Create New. Interface wan1, External IP 203.0.113.10, Map to 10.10.10.10, Port Forwarding TCP 80 → 80.

2. **Create the inbound policy wan1 → dmz with the VIP as destination**

   Firewall Policy > Create New: Incoming wan1, Outgoing dmz, Source all, Destination = your VIP, Service HTTP, ACCEPT, NAT off.

3. **Internet-User reaches http://203.0.113.10**

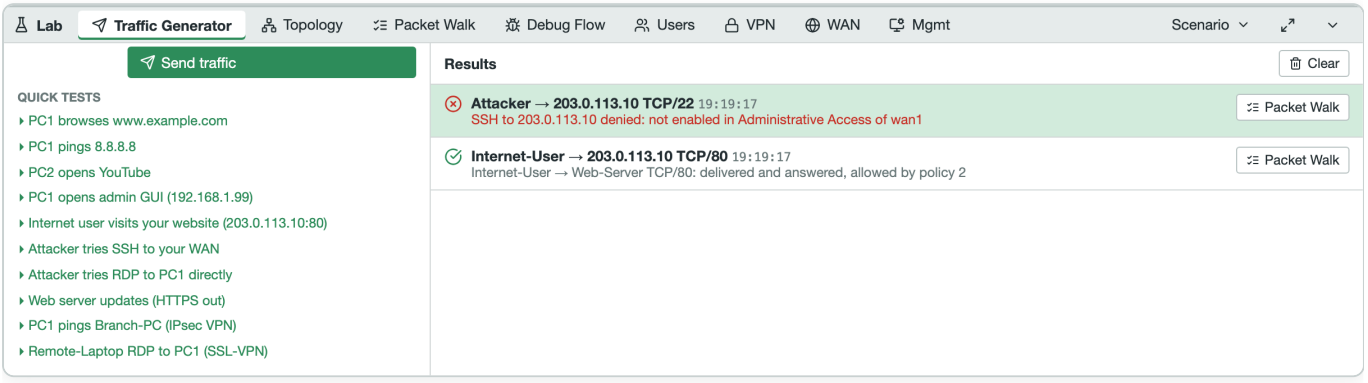
   Traffic Generator: click the quick test "Internet user visits your website" (Internet-User → 203.0.113.10, HTTP). The Packet Walk shows the DNAT step.

4. **SSH to 203.0.113.10 from the internet is still blocked**

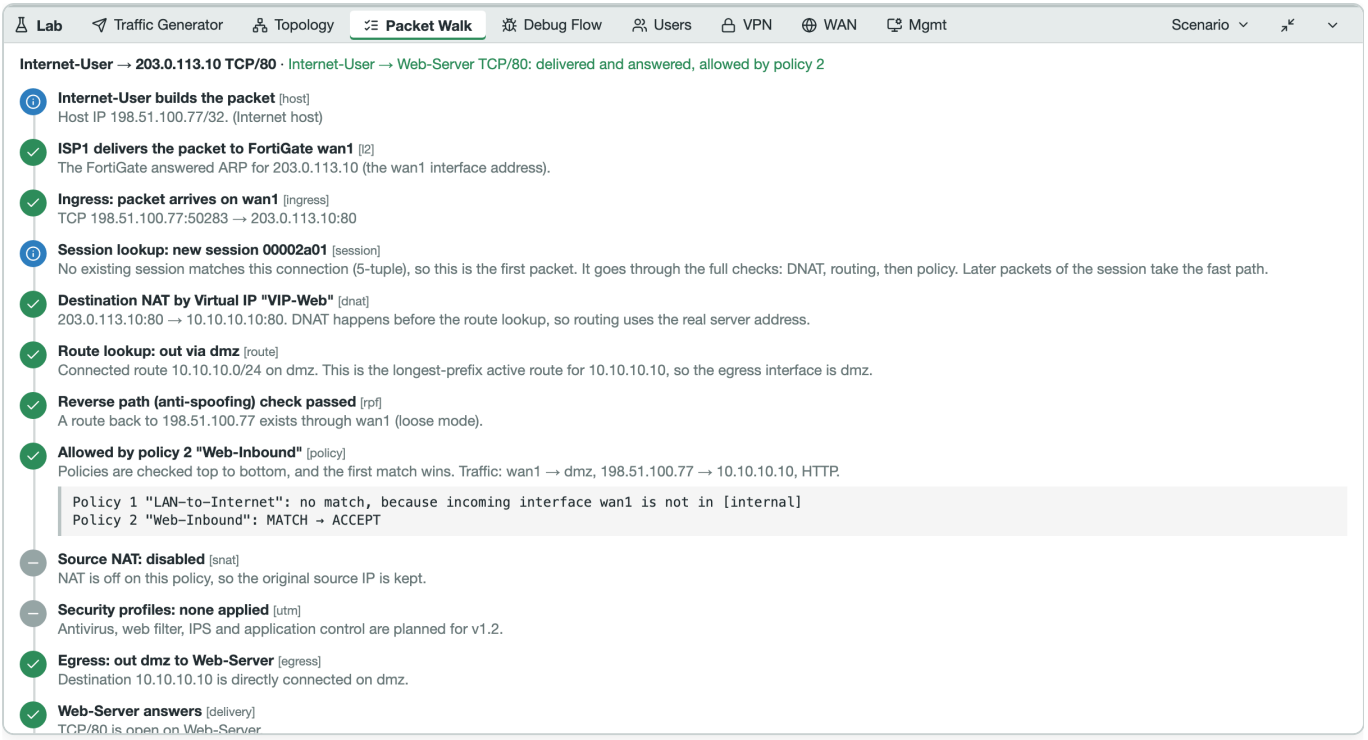
   Traffic Generator: click the quick test "Attacker tries SSH to your WAN" (Attacker → 203.0.113.10, SSH TCP/22). It must be blocked: only port 80 is forwarded to the web server, and SSH is not allowed on wan1.

Screenshots

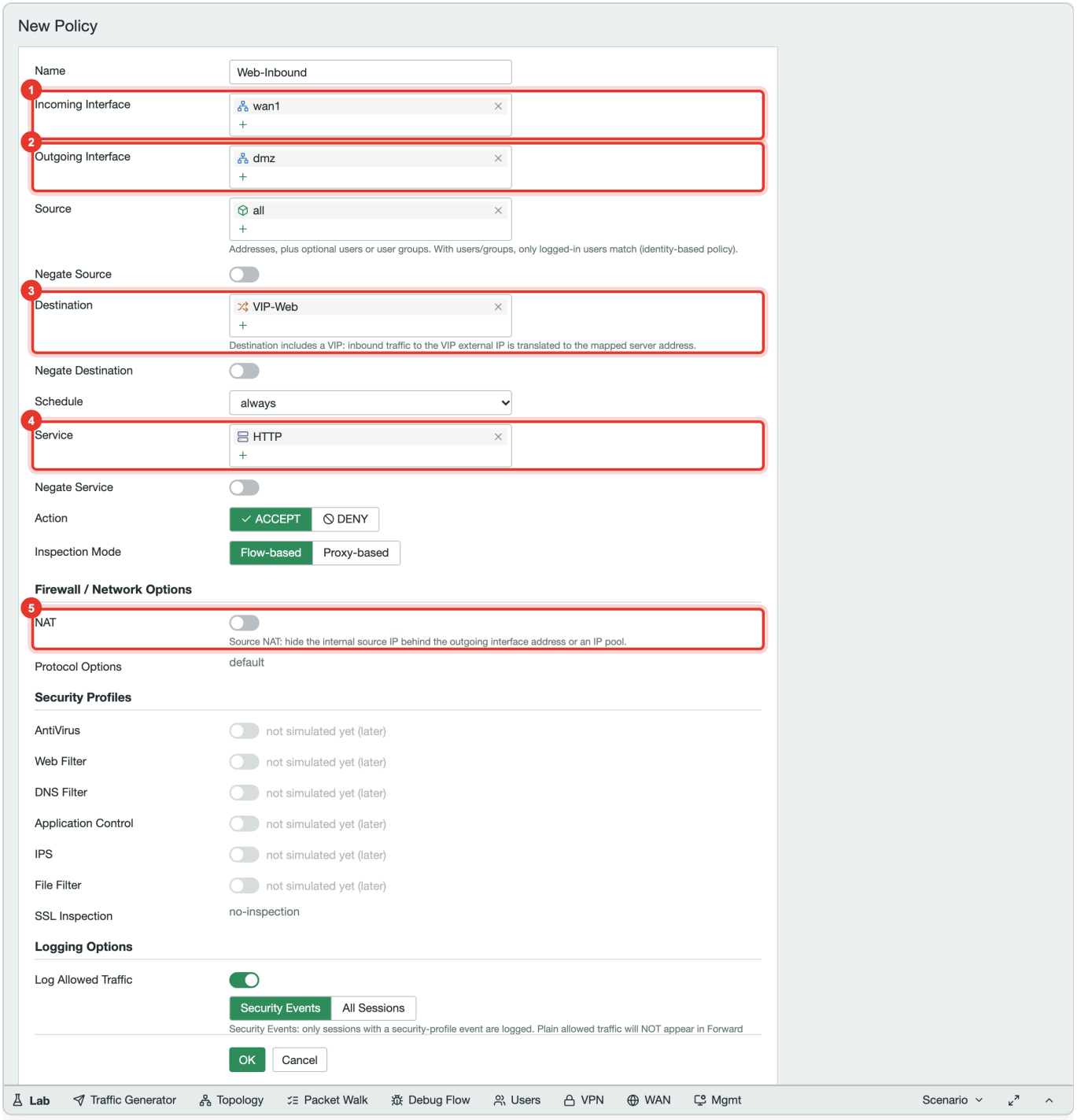
Numbered red markers show where to click or what to look at.



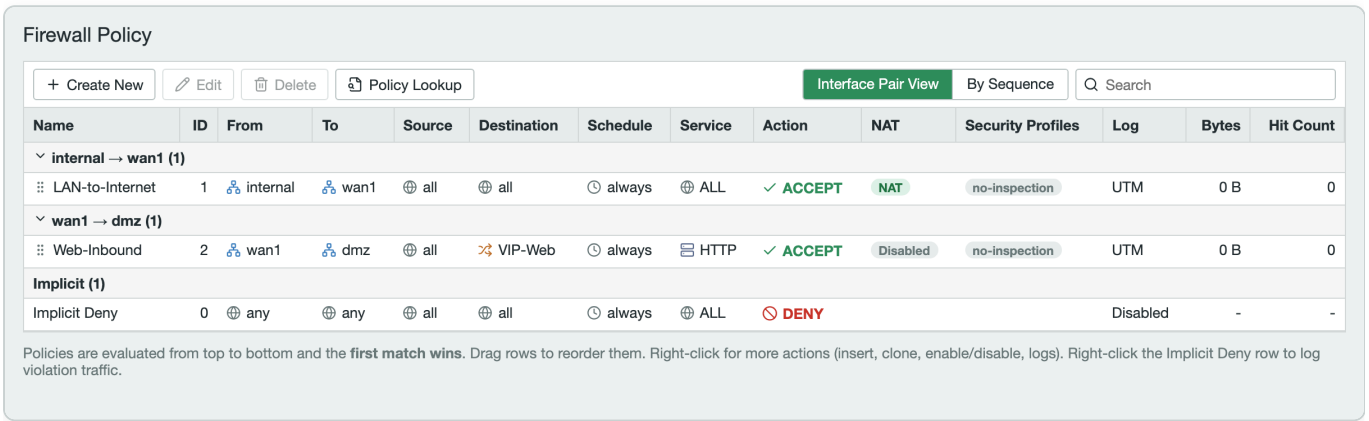
Attacker


Lab
Traffic Generator
Topology
Packet Walk
Debug Flow
Users
VPN
WAN
Mgmt
Scenario
Internet-User → 203.0.113.10 TCP/80 · Internet-User → Web-Server TCP/80: delivered and answered, allowed by policy 2
Internet-User builds the packet [host]
Host IP 198.51.100.77/32. (Internet host)
ISP1 delivers the packet to FortiGate wan1 [l2]
The FortiGate answered ARP for 203.0.113.10 (the wan1 interface address).
Ingress: packet arrives on wan1 [ingress]
TCP 198.51.100.77:50283 → 203.0.113.10:80
Session lookup: new session 00002a01 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.
Destination NAT by Virtual IP "VIP-Web" [dnat]
203.0.113.10:80 → 10.10.10.10:80. DNAT happens before the route lookup, so routing uses the real server address.
Route lookup: out via dmz [route]
Connected route 10.10.10.0/24 on dmz. This is the longest-prefix active route for 10.10.10.10, so the egress interface is dmz.
Reverse path (anti-spoofing) check passed [rpf]
A route back to 198.51.100.77 exists through wan1 (loose mode).
Allowed by policy 2 "Web-Inbound" [policy]
Policies are checked top to bottom, and the first match wins. Traffic: wan1 → dmz, 198.51.100.77 → 10.10.10.10, HTTP.
Policy 1 "LAN-to-Internet": no match, because incoming interface wan1 is not in [internal]
Policy 2 "Web-Inbound": MATCH → ACCEPT
Source NAT: disabled [snat]
NAT is off on this policy, so the original source IP is kept.
Security profiles: none applied [utm]
Antivirus, web filter, IPS and application control are planned for v1.2.
Egress: out dmz to Web-Server [egress]
Destination 10.10.10.10 is directly connected on dmz.
Web-Server answers [delivery]
TCP/80 is open on Web-Server

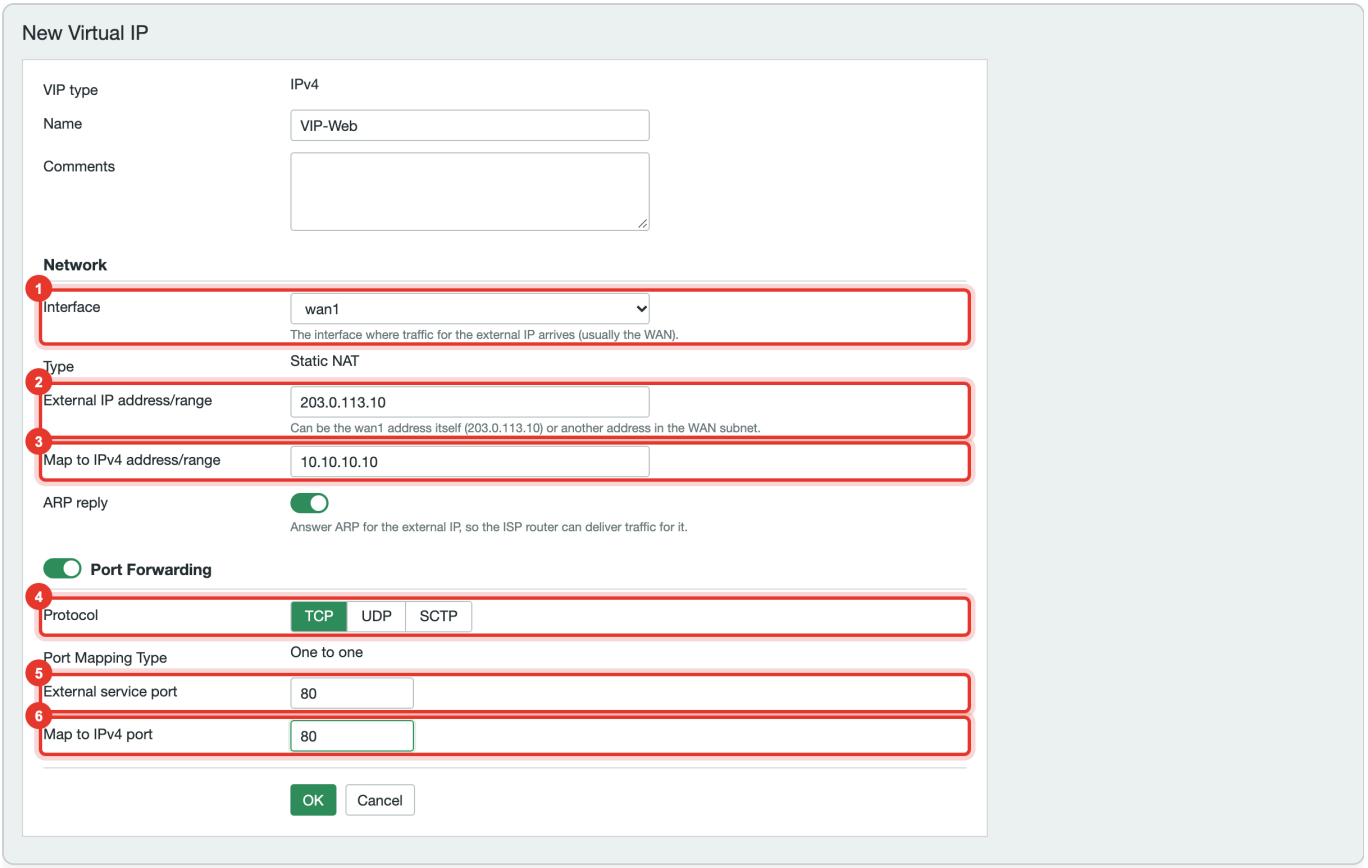
DNAT Walk



Inbound Policy



Policy List



Vip Form

Next: Class 6: Rule order matters